Shashanka Luitel

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Education

Kathmandu University — Dhulikhel, Nepal
Bachelor of Computer Engineering — *Dec. 2020 – Aug 2025*
- GPA: **3.69 / 4.0**

DAV College — Lalitpur, Nepal
+2 Science, NEB — *2018 – 2020*
- CGPA: **3.76 / 4.0**

Experience

Software Engineer — Apr. 2025 – Dec 2025
Citytech Group Pvt. Ltd. — *Kamaladi, Kathmandu*
- Contributed to the development and enhancement of remitPulse, a production-level fintech admin portal used for remittance and compliance workflows
- Implemented Redux for global state management and handling asynchronous API calls
- Analyzed and modified existing stored procedures, gaining hands-on experience with MS SQL queries
- Tested and validated APIs using tools such as Postman and Swagger
- Actively participated in Agile workflows including daily standups, sprint planning, tasks ticketing, git branching, pull requests, and peer code reviews via GitHub

Projects

MindSense - Machine Learning-Based Mental Health Score Prediction | GitHub | Demo
Tech Stack: Next.js, TypeScript, FastAPI, Python, Pandas, Scikit-learn
- Developed a full-stack machine learning application that predicts mental health scores based on users' digital habits, lifestyle, and academic factors
- Performed data preprocessing before training the machine learning model, and integrated it with a FastAPI backend for real-time predictions

TripSync - Multi-Agent AI Trip Planner | GitHub | Demo
Tech Stack: Next.js, TypeScript, FastAPI, Python, LangChain, LangGraph, PostgreSQL, Groq, Tavily
- Developed an AI-powered trip planner that generates personalized, budget-aware itineraries using a multi-agent workflow
- Integrated LangGraph with web search and travel APIs to gather relevant information for itinerary planning
- Implemented PostgreSQL-backed conversation history, trip history, and PDF itinerary exports

Trek Diaries - Social Media Platform for Trekking Enthusiasts | GitHub | Demo
Tech Stack: Next.js, TailwindCSS, NeonDB, DrizzleORM, Cloudinary, NextAuth, Expo, Redis, Algolia
- Built a trekking-focused social platform enabling users to share posts and interact through likes and comments
- Integrated a recommendation system to personalize content and enhance user experience
- **Mobile App** : GitHub

Battleship - Multiplayer Strategic Game | GitHub | Demo
Tech Stack: React.js, Node.js, MongoDB, Socket.io
- Developed a browser-based multiplayer tactical game where players command fleets of warships
- Implemented turn-based gameplay with logic to track guesses, hits, misses, and fleet destruction conditions

Flight Delay Data Visualization | GitHub | Demo
Tech Stack: Streamlit, Python
- Developed a comprehensive data mining and visualization dashboard to analyze historical flight data
- Processed flight delays, cancellation reasons, and temporal trends to identify patterns and provide insights

Maze Runner - 2D Simulation Game | GitHub
Tech Stack: Python
- Developed a 2D car simulation game using Python, Pygame, and PyOpenGL
- Players navigate a car through a maze to reach the finish line

Research Work

Digital Health in Nepal: Past, Current and Future Scenarios. Survey paper, Department of Computer Science and Engineering, Kathmandu University. Link

Ethics of Modern Media Provocation. Link

Technical Skills

Programming Languages: JavaScript, Python, C/C++, SQL, HTML, CSS

Frontend: React.js, Next.js, React Native, Expo, Redux and Redux Toolkit, Zustand, TailwindCSS, ShadcnUI

Backend: Node.js, Express.js, FastAPI, Flask, Socket.Io

Database: MySQL, PostgreSQL, MongoDB, Redis, Firebase, Appwrite

Machine Learning: Scikit-learn, Pandas, NumPy, Seaborn, Joblib, Data Preprocessing, Model Evaluation

Tools: Git and GitHub, Unity, Docker, Swagger, Postman, Streamlit, DataGrip, Android Studio

Soft Skills

Communication, Collaboration, Presentation, Problem-solving, Teamwork, Time Management

Extracurricular Activities

Developer Lead Aug. 2024 – Dec 2024

IT MEET 2024

- Leading a team of developers to build and maintain a fully responsive and dynamic website of IT MEET 2024, which is an annual, national-level event organized by Kathmandu University Computer Club (KUCC)
- Collaborating with the design team to ensure seamless UI/UX across all devices

Beta Microsoft Learn Student Ambassador Nov. 2023 – Mar 2025

MICROSOFT

- Conducted workshops and events in multiple institutions to empower students to cultivate early coding habits
- Received training and resources from Microsoft to enhance technical skills and leadership abilities

Designer Sub-Lead Sep. 2023 – Oct 2023

KU HACKFEST 2023

- Designed visual assets for the event, ensuring alignment with the overall theme and objectives
- Collaborated with other teams to maintain a consistent and cohesive event presentation

Certifications

Basic concept of AI and ML (Link)

Neural Network and Deep Learning (Link)

Principles of UX/UI Design (Link)

React Native (Link)

Languages

Nepali [Native] **English** [Fluent] **Hindi** [Fluent] **German** [Learning] **French** [Learning]